

Public-Data and Source-Port Audit for the Tau Core Quantum Terminal

Technical supplement to Paper VI

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Abstract

This supplement preserves the full candidate census, negative controls, source-port audits, finite countermodels and readiness scoring underlying the compact public-data verdict in Paper VI. It does not add a new Tau detection. Its central result is an identifiability no-go: no audited public carrier simultaneously supplies an independent once-counted action-side observable, an Uhlmann-capable endpoint record and the required recovery and holonomy ledger. All scores are capability or eligibility scores, not signal significances.

Dependency. Equation and theorem references prefixed by the main-paper numbering refer to Paper VI. This supplement owns only the detailed audit ledger and its finite controls; it does not strengthen the main paper’s conditional claims.

1 Detailed audit

1.1 Finite public-carrier census

To prevent repeated searches in non-identifying datasets, we froze eight public candidates and applied five gates: same carrier, an x_Q -capable state distance, an action-like observable, an independently once-counted x_A , and errors plus recovery information. Table 1 summarizes the result. “Partial” means that the required observable exists only as a model-dependent or same-pipeline proxy.

Table 1: Public-carrier census. No candidate passes the independent- x_A gate.

Carrier/data	same	x_Q	action-like	independent x_A	errors/recovery
Tan superconducting qubit	yes	yes	yes	no	partial
Yu NV centre	yes	yes	no	no	partial
Kim black phosphorus	yes	yes	yes	no	partial
Zheng non-Abelian circuit	yes	yes	yes	no	partial
Pires NMR open QSL	yes	yes	yes	no	partial
Ness trapped atom QSL	yes	yes	yes	no	yes
KIT mitigated tomography	yes	yes	no	no	yes
Photonic overlap data	yes	yes	no	no	partial

The strongest standard-QOR control is the trapped-atom dataset of Ness et al. [1]: it publishes two-time overlaps together with mean energy, energy variance, timing and an associated data archive. It can test the ordinary speed-limit side, but energy moments are not an independently defined morphological action separation. We audited all 53 published Figure 2 overlap points against the larger of the interpolated Mandelstam–Tamm and Margolus–Levitin lower limits. Pointwise central values occasionally fall below the fitted limit, but after combining the measured lower and bound upper uncertainties in quadrature the largest one-sided deviation is

only 1.4465σ ; no point exceeds 1.96σ . Thus the public control is consistent with the standard QSL and supplies no anomalous Tau residual.

The strongest spectral-response candidate is the public black-phosphorus package of Kim et al. [2], which contains ARPES dispersion and QGT products. Its x_A map is not source-frozen, and the QGT and candidate spectral response share ARPES/pseudospin processing. The non-Abelian superconducting-circuit [3] and open-NMR QSL [4] candidates fail for the analogous same-pipeline reason.

There is also a structural obstruction to simply declaring spectral curvature to be x_A . For the generic two-band family

$$H(q) = r(q) \mathbf{n}(q) \cdot \boldsymbol{\sigma}, \quad r(q) = 1 + bq^2, \quad \mathbf{n}(q) = (\sin aq, 0, \cos aq), \quad (1)$$

the ground-state quantum metric and positive-band curvature at $q = 0$ are

$$g_{qq}(0) = \frac{a^2}{4}, \quad E''_+(0) = 2b. \quad (2)$$

Holding a fixed while varying b gives the same QGT and different spectral curvature; holding b fixed while varying a gives the same spectrum and different QGT. Hence neither quantity determines the other without an additional source-owned law tying radial spectral data to angular state data. This countermodel rules out relabelling an ARPES band curvature as an independent morphological action record.

The same obstruction survives when a public calculation supplies separate energy- and fidelity-susceptibility estimators, as in the open PMR-QMC packet of Ezzell et al. [5]. For a nondegenerate ground state of $H(\lambda) = H_0 + \lambda V$, standard perturbation theory gives

$$\chi_E = -\partial_\lambda^2 E_0 = 2 \sum_{n>0} \frac{|V_{n0}|^2}{\Delta_n}, \quad \chi_F = \sum_{n>0} \frac{|V_{n0}|^2}{\Delta_n^2}. \quad (3)$$

They are different spectral moments. Already for one excited level, $\chi_E/\chi_F = 2\Delta$, so no parameter-independent universal scale converts one into the other. Independent estimation is therefore necessary but not sufficient: identifying an action-side x_A requires a source-owned rule that fixes the gap weighting and normalization without using the fidelity data. The public PMR-QMC package is useful as a computational control of the two moments, not as an occupied physical carrier or a test of Eq. (41).

Consequently the census has zero full passes. Ness 2021 is retained only as a standard QOR/recovery control. Kim 2025 may be reopened only after an action/work/response-curvature map is defined without using the state or QGT data that produce x_Q . This is the stopping verdict for the present public inventory; adding another tomography, metric or Berry estimator cannot reduce the finite discriminator blocker. The complete Ness tables, provenance, uncertainty audit and two-band countermodel are included in the reproducibility package.

1.2 Operational boundary for the action-side observable

The preceding audits permit a finite exhaustion statement for the standard public-observable routes. Let $\mathfrak{D}_Q$ be the algebra generated by state tomography, Hamiltonian spectroscopy, quantum-geometric response and QOR timing on the selected carrier. Terminally equivalent completions can agree on all of $\mathfrak{D}_Q$ while assigning

$$x_A^{(0)} = x_Q, \quad x_A^{(\lambda)} = x_Q + \lambda x_Q^2, \quad \lambda > 0. \quad (4)$$

Therefore no statistic measurable only in $\mathfrak{D}_Q$ identifies finite A8b. The concrete candidate reductions fail for complementary reasons:

(O1) QGT, fidelity and tomography are functions of $\mathfrak{D}_Q$ and are circular as action-side estimators;

- (O2) reversible quasistatic work vanishes for a fixed-gap rotating ground state although its endpoint Uhlmann chord is nonzero;
- (O3) band curvature is independent of the eigenstate QGT in the two-band countermodel above;
- (O4) energy susceptibility and fidelity susceptibility are inequivalent spectral moments by Eq. (3).

Operational x_A no-go. Within the frozen public inventory and the observable algebra $\mathfrak{O}_Q$, there is no operationally identified independent estimator of $x_A = \mathcal{A}_X^{PC}/A_*$. The common-Gram equality remains a conditional cross-domain prediction, not an executable public-data test.

This no-go does not make the prediction unfalsifiable in principle. It fixes the minimum admissible reopening packet. One must independently measure a source/body-side local stiffness $K_A(q)$, freeze its action unit and continuation law before examining withheld quantum endpoints, and then test the blind predictions

$$\widehat{x}_A(q_i, q_j) \stackrel{\text{blind}}{=} 2(1 - \sqrt{F(\rho_i, \rho_j)}) \quad (5)$$

on non-collinear pairs and closed triangles. Triangle inconsistency would detect unmodelled holonomy or finite-response corrections. Relative distance ratios can reject nonlinear laws before A_* is known, but full A8b additionally requires the common absolute action unit. No located public dataset contains this independent stiffness-plus-withheld-endpoint packet.

Local stiffness alone is still insufficient. Consider two amplitude-plane embeddings

$$W_0(q) = (q, 0), \quad W_\kappa(q) = \frac{1}{\kappa}(\sin \kappa q, 1 - \cos \kappa q). \quad (6)$$

Both have the identical local pullback metric $\|\partial_q W\|^2 = 1$, but for an endpoint interval Δq their chords are

$$d_0 = |\Delta q|, \quad d_\kappa = \frac{2}{|\kappa|} \left| \sin \frac{\kappa \Delta q}{2} \right|, \quad (7)$$

which differ at finite separation. Thus K_A determines only intrinsic infinitesimal length. A blind finite certificate must additionally freeze the connector's extrinsic continuation, or an equivalent amplitude lift and connection, and test its holonomy. The minimal reopening packet is therefore

$$\boxed{\{K_A, C_q, \nabla^C, A_*, \text{withheld endpoints and loops}\}}. \quad (8)$$

On a simply connected tested domain, flat right-unitary-basic transport makes the integrated lift path independent and permits Eq. (5). Nontrivial descriptor-visible holonomy must be retained as physical path data rather than quotiented away.

Even all pairwise endpoint distances do not determine that loop datum. For the three pure qubit states

$$|\psi_0\rangle = (1, 0), \quad |\psi_1\rangle = (\cos \theta, \sin \theta), \quad |\psi_2^\pm\rangle = (\cos \theta, e^{\pm i\phi} \sin \theta), \quad (9)$$

the $+$ and $-$ packets have identical three pairwise fidelities. Their gauge-invariant Bargmann products

$$\mathcal{B}_\pm = \langle \psi_0 | \psi_1 \rangle \langle \psi_1 | \psi_2^\pm \rangle \langle \psi_2^\pm | \psi_0 \rangle \quad (10)$$

are complex conjugates and carry opposite nonzero phases. Pairwise chords therefore cannot certify the connector's phase transport. At least one closed triangular loop witness is mandatory in addition to non-collinear endpoint tests. For mixed states the corresponding object is the declared Uhlmann transport/holonomy, not an arbitrary purification phase.

1.3 Alternative physical sources

The operational no-go above excludes reconstruction of x_A from the frozen standard quantum-observable algebra; it does not exclude a different physical measurement source. We therefore performed a finite same-carrier audit of six experiments that realize complementary parts of the blind packet. Batalhão et al. directly reconstructed a quantum work distribution in an NMR spin system [6]. Cottet et al. combined microwave work extraction with full state tomography in a superconducting quantum Maxwell-demon experiment [7]. Mixed-state geometric phase was measured interferometrically in NMR [8], and an Uhlmann phase was implemented with superconducting qubits [9]. Finally, an Xmon experiment combined a closed geometric gate with process tomography [10]. Naghiloo et al. additionally tracked individual transmon trajectories and assigned heat and work along the same reconstructed state path [11].

These results establish the separate experimental availability of all three measurement legs,

$$\text{work/action response,} \quad \text{endpoint state geometry,} \quad \text{closed-loop phase or holonomy.} \quad (11)$$

They do not yet form one source-frozen certificate. No audited publication simultaneously supplies all three legs on the same physical carrier, an independent map from the measured work-like observable to the body action, and a reusable numerical archive sufficient for the full test. In particular, thermodynamic work and imposed control-pulse area are standard terminal observables; neither may be renamed $\mathcal{A}_X^{PC}$ without a source-owned calibration theorem. The Naghiloo packet is the strongest trajectory-resolved candidate, but its work is computed from monitored $\rho(t)$ and known $H(t)$; it therefore adds no independent action coordinate, reports no closed Bargmann/Uhlmann witness, and has no located reusable raw-trajectory archive. Its frozen publication-level readiness score is therefore 6/13. The reported 20 ns sampling, approximately 10^4 trajectories, measurement efficiency 0.35, heat–feedback correlation -0.68 , and feedback/open-loop regression slopes $0.437/0.006$ support strong first-law and feedback controls. They do not license the finite A8b, holonomy or QOR scores.

The Cottet packet plays a complementary and stronger instrumental role. Extracted work is measured directly as an input–output microwave-power difference and checked against an independent qubit-energy measurement and full cavity-memory tomography. Its readiness score is 7/13. This is an independent measurement instrument, not yet an information-independent terminal coordinate. Reported controls include informed/ignorant memory occupations $\bar{n} = 9/0$, residual qubit excitation 0.027 ± 0.010 , and memory entropies 1.0 ± 0.05 versus 1.2 ± 0.1 . It still does not supply a source-frozen map from this terminal work to $\mathcal{A}_X^{PC}/A_*$ or a closed holonomy. It is therefore the primary directly measured energy control, while the Naghiloo packet remains the trajectory control; their records may not be stitched into one Tau certificate.

The distinction follows directly from input–output theory. With frozen input field and coupling controls c , the measured output satisfies

$$b_{\text{out}} = b_{\text{in}} - \sqrt{\gamma_b} \sigma_-, \quad \bar{P}_{\text{out}} = F(\rho_S, c). \quad (12)$$

Consequently $D\bar{P}_{\text{out}} = D_\rho F D\rho_S$ and stacking mean output power with complete qubit tomography does not increase the conditional descriptor rank. The power detector is physically independent and provides an excellent cross-check, but its mean record is not information-novel once (ρ_S, c) is known.

There is a precise qualification for fluctuations. Multi-time output moments need not factor through an instantaneous $\rho_S(t)$ alone: they also depend on the dynamical generator $\mathcal{L}$, coupling operators, input-field state and controls. Under a frozen Markov input–output model, however, quantum regression gives schematically

$$\langle O_n(t_n) \cdots O_1(t_1) \rangle = \mathcal{F}_n(\rho_S(0), \mathcal{L}, L_b, \rho_{\text{in}}, c). \quad (13)$$

Correlations may therefore be novel relative to endpoint or instantaneous tomography while remaining redundant relative to independently frozen process tomography. A residual surviving

the latter test diagnoses an incomplete standard process model, not automatically the Tau body action. The Cottet publication-level packet reports averaged power and no reusable single-shot archive sufficient for this residual test, so no fluctuation score is licensed.

We therefore performed a finite direct-public-data audit. The Wang longitudinal-readout Zenodo packet contains three pairs of prepared-state single-shot IQ arrays with approximately 2.94×10^4 records per class. Projection onto the frozen class-centroid axis and classification at its midpoint gives accuracies 0.9951, 0.9956 and 0.9980, with pooled d' values 7.32, 6.34 and 8.63. These are reproducible standard readout controls. They are integrated endpoint samples, not time-resolved trajectories, and supply neither a process residual, holonomy nor body action. The public Murch thermal-readout packet contains accumulated probabilities and histograms rather than individual trajectories. Fichoux reports the strongest architecture: time-resolved records, a frozen stochastic master equation and independent terminal tomography on one carrier. Its raw records are available only on request; public movies are not a reusable numerical archive. Under the declared direct-public rule, zero located packets pass the six-part process-residual gate. This closes the current public-data search at the Tau-scoring level while retaining Wang as a pipeline control and Fichoux as the leading acquisition target.

A separate public optical packet provides a useful same-experiment control without repairing this identifiability failure. Wen et al. reconstructed single-photon amplitudes over more than 1.4 million paths and released the figure tables under CC0 [12]. Across the 100 published action bins, a weighted linear fit of path amplitude against classical action gives slope -0.0287 ± 0.0417 , consistent with the equal-magnitude postulate. The measured phase follows the reported action prediction with RMS residual 0.0218 in units of π . These are strong standard path-integral controls. They are not a Tau score: path length and action are supplied as separate aggregate tables, not as joint path-level (M, S, A, θ) records. Hence a conditional novelty test such as $I(A; M | S) > 0$ is not identifiable, and the optical action cannot be renamed as $\mathcal{A}_X^{PC}$.

The raw camera matrices nevertheless allow one bounded follow-up diagnostic. Using the published polarization differences, probability normalization and the quadratic vertex of the independent Fig. S1C sample, we reconstruct 17 radial step-kernel classes and enumerate the full $17^5 = 1,419,857$ path family. The phase reduction reproduces the Fig. S1C sample to RMS 0.0120π after one common phase offset. Relative to scalar total standard action, five frozen path descriptors add $\Delta R^2 = 0.1758$ to the phase residual regression. The generated paths, however, share the same 17 measured kernel classes. A permutation test at that kernel level gives $p = 0.144$; there is therefore no resolved morphology increment on the validated phase leg. The image-level amplitude reduction gives a larger apparent increment but fails the independent Fig. S1C amplitude check and is excluded. This is a standard factorized-kernel diagnostic, not evidence for an independent Tau body action or a Tau-specific quantum signal.

The released Figs. S2C–S2D permit a corrected amplitude-side reconstruction without the failed image-level normalization. Their 17 normalized radial means have only 0.863% coefficient of variation, yet the corresponding path products retain $\Delta R^2 = 0.4962$ beyond scalar total action. A permutation of the 17 shared measured kernel classes gives $p = 0.004$. Propagating the histogram-derived standard errors through 9,999 draws under a constant- amplitude null gives $p = 0.0009$, and replacing any single radial class by its action-trend prediction leaves $\Delta R^2 \geq 0.2936$. Thus the amplitude structure is reproducible from the authors' published reduction and is not a single-class artifact. It remains a standard optical-kernel morphology signal, not a Tau endpoint: all paths reuse the same 17 measurements and the standard optical action is not an independently reconstructed $\mathcal{A}_X^{PC}$.

To avoid interpreting this five-descriptor fit after the fact, we next freeze one theory-motivated candidate before comparison with the controls. The candidate is the Dirichlet energy of the path-increment field,

$$M_{\text{curv}}(\gamma) = \sum_k (\Delta x_{k+1} - \Delta x_k)^2. \quad (14)$$

This is only a conditional Tau descriptor: the body program selects a primitive Dirichlet form,

whereas applying it to the increment field requires a separately typed, body-derived edge carrier that is not proved here. With scalar total action retained as the baseline, Eq. (14) adds only $\Delta R^2 = 0.000405$. A permutation of the 17 shared radial kernel classes gives $p = 0.543$, and the candidate ranks fourth among the five single-descriptor additions. Ordinary path length alone gives $\Delta R^2 = 0.4826$. Thus the robust published-amplitude result is dominated by a standard accumulated-step descriptor; the frozen curvature candidate is not detected. This negative result narrows the admissible Tau morphology family and prevents the broad five-variable signal from being promoted to a Tau-specific effect.

An independent aggregate check is supplied by the released Fig. S2B table. It contains 17 measured radial phase points with standard deviations and a 200-point theoretical curve. Linear interpolation of the theory onto the measurement grid gives an RMS residual of 0.01233π ; all points lie within one reported standard deviation, with maximum absolute standardized residual 0.4873. This strengthens the standard phase reconstruction but not the Tau identifiability claim: Fig. S2B contains no same-index complex amplitude, recovery-process record or independently reconstructed body action. It therefore remains ineligible for the primary Tau score.

This gives a positive but conditional reopening architecture. On one programmable qubit-ancilla carrier, reconstruct the characteristic function of work by Ramsey interferometry, independently tomograph withheld endpoints, and close at least one Bargmann or Uhlmann loop. Freeze the action map and A_* before reading the endpoint and loop data. The resulting packet would test both the finite magnitude identity and connector transport without cross-platform stitching. The audit therefore replaces an open-ended source search by one finite design target; it is a feasibility result, not evidence that Nature realizes finite A8b.

The three legs are also mathematically compatible on one explicit carrier. Consider the closed qubit family

$$H(q) = \frac{\Delta_0(1 + m \cos q)}{2} \mathbf{n}(q) \cdot \boldsymbol{\sigma}, \quad \mathbf{n}(q) = (\sin \theta \cos q, \sin \theta \sin q, \cos \theta), \quad (15)$$

with Gibbs records $\rho(q) = e^{-\beta H(q)} / \text{Tr } e^{-\beta H(q)}$. On this same path, a sudden step has the standard work characteristic function

$$G_{q \rightarrow q'}(u) = \text{Tr} \left[e^{iuH(q')} e^{-iuH(q)} \rho(q) \right], \quad (16)$$

tomography gives the Uhlmann chord $x_Q = 2(1 - \sqrt{F(\rho(q), \rho(q'))})$, and discrete Uhlmann parallel transport around $q \in [0, 2\pi]$ gives a closed-loop phase. For the frozen control $(\beta, \Delta_0, m, \theta) = (1.3, 1, 0.35, 1)$ on 65 points, the audit finds a nonconstant $G(u)$, nonzero step chords and loop phase $\Phi_U = 0.13427$ radians; the final transport-unitarity defect is below 8×10^{-16} .

Joint-carrier compatibility result. Work-characteristic interferometry, endpoint state geometry and Uhlmann loop transport can therefore be defined consistently on one Hamiltonian family without combining different carriers. The construction does not prove $x_A = x_Q$: Eq. (16) is a terminal thermodynamic work statistic, and the independent map from this statistic to $\mathcal{A}_X^{PC}/A_*$ remains absent. Its role is to remove protocol incompatibility as a blocker and isolate the remaining source-law question.

There is, however, a finite obstruction to using ordinary dissipated work as that missing source law. For a sudden step between Gibbs states, direct substitution of $\log \rho(q) = -\beta H(q) - \log Z(q)$ gives

$$\beta(\langle W \rangle - \Delta F) = D(\rho(q) \| \rho(q')). \quad (17)$$

Thus dissipated work induces relative-entropy/Bogoliubov-Kubo-Mori geometry, whereas the finite A8b target uses the Uhlmann/Bures chord. These geometries are not related by one universal constant. On the frozen loop above, the stepwise ratio $D(\rho(q) \| \rho(q'))/x_Q(q, q')$ ranges from 2.1171 to 2.4882.

Ordinary-work bridge no-go. Even on the joint carrier, no path-independent common gain converts ordinary dissipated work into the finite Uhlmann chord. Work interferometry remains a valuable independent control and may constrain a richer source map, but the identification $\beta W_{\text{diss}} = x_A = x_Q$ is rejected. A successful bridge must instead descend the already defined once-counted body action, or derive a nontrivial state-dependent transformation before endpoint data are read.

The inherited body construction suggests a different, noncircular work observable. Let z be a source-side occupied defect coordinate and let its once-counted positive body action be

$$\mathcal{A}_X^{PC}(z) = \frac{1}{2} \langle z, K_A(z) z \rangle, \quad K_A(z) > 0. \quad (18)$$

Its conjugate body-work one-form is not the terminal qubit energy change but

$$\alpha_B := d\mathcal{A}_X^{PC}, \quad x_A(z) = \frac{1}{A_*} \int_0^z \alpha_B = \frac{\mathcal{A}_X^{PC}(z)}{A_*}, \quad (19)$$

where the zero-defect reference, physical path family and A_* are frozen before state tomography. Exactness makes the integral path independent on the tested simply connected body chart; failure of closure is itself a rejection of this minimal bridge and must not be repaired by endpoint fitting.

Conditional body-work bridge. An independent generalized-force reconstruction of α_B , including its closure and normalization ledger, supplies the missing action ordinate without using ρ , fidelity or QGT. Equation (41) can then be tested blindly against withheld endpoint tomography and loop holonomy. This is an operational realization of the inherited once-only body-energy transgression, not a proof that a presently published qubit experiment has measured it.

Resonator-factorization no-go. A readout resonator is not automatically a second physical axis. Let c denote frozen controls and suppose its reported response satisfies

$$R_{\text{res}} = F(\rho, c). \quad (20)$$

Then $DR_{\text{res}} = D_\rho F D\rho$, so $\ker D\rho \subseteq \ker D(\rho, R_{\text{res}})$ and stacking the resonator record with state tomography cannot increase rank beyond the state map. Equivalently, the response contains no conditional information about a hidden action coordinate once (ρ, c) is known. Thus dispersive cavity shifts used to reconstruct the qubit state, and dynamical responses designed to reconstruct the QGT, cannot also serve as an independent x_A ordinate.

A finite seven-candidate audit separates two questions. Dispersive readout, measurement-instrument backaction and QGT-response protocols do not supply an independent action axis [10, 13–17]. Calorimetric work measurement is the one audited architecture that can add an independent energy-exchange record [18], but the cited construction does not provide a public same-carrier endpoint/holonomy packet or a frozen map from ordinary energy exchange to $\mathcal{A}_X^{PC}/A_*$. Thus independent measurement rank is necessary but not sufficient for an independent x_A . The positive requirement is a separately calibrated source-port force, impedance, admittance or calorimetric backaction whose response does not factor through ρ or the QGT, followed by a body-action map frozen before tomography.

Generalized-force factorization no-go. The phrase “generalized force” does not by itself meet this requirement. For a standard controlled Hamiltonian family $H(\lambda)$, the measured force is

$$\mathcal{M}_\mu = -\text{Tr}(\rho \partial_\mu H). \quad (21)$$

Once ρ , H and the control calibration are frozen, $\mathcal{M}_\mu = F_\mu(\rho, H, \partial H)$ and its Jacobian row lies in the span of those inputs. Slow-drive corrections can expose Berry curvature and other quantum-geometric response coefficients; this makes generalized-force experiments valuable independent

instrumental checks of quantum geometry, but not an information-independent ordinate for x_A . Using the same response both to reconstruct the QGT and to test $x_A = x_Q$ would therefore be circular [19, 20].

The shortcut reopens only if a separately calibrated residual

$$\xi_\mu := \mathcal{M}_\mu + \text{Tr}(\rho \partial_\mu H) \quad (22)$$

is reproducibly nonzero, survives conditioning on state, controls and instrument response, and is mapped to the once-counted body action with its unit frozen before endpoint tomography. A nonzero ξ_μ would first diagnose an incomplete standard source-port model; it would become a Tau test only after that independent body-action bridge is established. Thus the standard generalized-force/QGT route is closed, while calorimetric or mechanical source-port architectures remain conditionally open.

Bounded public source-port verdict. We finally audited the strongest public thermal and mechanical alternatives rather than extending the candidate census indefinitely. Gunyhó et al. publish a large single-shot thermal-detector archive, but the bolometer absorbs the state-dependent resonator readout power; it is a physically distinct detector, not a conditionally independent source/body ordinate [21]. Pulsed optomechanical tomography reconstructs mechanical phase space with calibrated optical pulses, while collective backaction-evading measurements establish mechanical tomography and force-sensing capability [22, 23]. Neither publication supplies a joint applied-force record surviving conditioning on the reconstructed state, a finite Uhlmann/QGT endpoint and a frozen map to $\mathcal{A}_X^{PC}/A_*$. The finite public-source-port audit therefore has zero eligible packets. This closes the present public-data-only acquisition class; Eq. (19) remains a future protocol or an internal source-law target, not an observed Tau signal.

Inherited CHDF body-work theorem. The internal source side is stronger than the public-data status. Let $L : X_B \rightarrow E_S$ be the established visible body map and let $H_B > 0$ be the frozen CHDF body Hessian. Exact constrained elimination gives

$$\mathcal{A}_Q(y) = \inf_{Lb=y} \frac{1}{2} \langle b, H_B b \rangle = \frac{1}{2} \langle y, K_{\text{leaf}} y \rangle, \quad K_{\text{leaf}} = (LH_B^{-1}L^*)^{-1} > 0. \quad (23)$$

Consequently the body-work form is not an additional ansatz:

$$\alpha_B = d\mathcal{A}_Q = \langle K_{\text{leaf}} y, dy \rangle, \quad d\alpha_B = 0, \quad \int_0^y \alpha_B = \mathcal{A}_Q(y). \quad (24)$$

It inherits the same parent action unit and is path independent on every simply connected quotient chart. The symbolic replay uses $k_R = 59/64 - \sqrt{3}/12$ and $k_T = 1 + \sqrt{3}/24$ and verifies positivity, closure and equality of straight and piecewise path integrals exactly.

This closes only the existence and normalization of the body-side one-form. Under the adopted PRRC-P1/PJR-P1 constitutive continuation, the protected body-relative role is carried into the post-body packet once, with no duplicate action or free relative scale. That working-completion statement is not a theorem that Nature selects PRRC-P1. More importantly, neither Eq. (23) nor PRRC-P1 constructs the observer amplitude connector required by finite A8b. The implication $\alpha_B \Rightarrow x_A = x_Q$ therefore remains forbidden: the former is now an inherited body theorem, while the latter is still the common-source connector claim to be proved or tested.

Finite-range realization and non-selection theorem. The body action above is quadratic and therefore unbounded, whereas the normalized Uhlmann chord

$$x_Q = 2(1 - \sqrt{F(\rho_+, \rho_-)}) \quad (25)$$

satisfies $0 \leq x_Q \leq 2$. Consequently a raw global identity $x_A = x_Q$ is impossible on the complete CHDF body. It can hold only on an occupied bounded sector, or after a separately derived bounded reparametrization of the action coordinate.

That bounded sector is nonempty and admits an exact construction. For $0 \leq x \leq 2$, define

$$\rho_{\pm}(x) = \frac{1}{2} (I \pm r(x)\sigma_z), \quad r(x)^2 = x - \frac{x^2}{4}. \quad (26)$$

The states are positive and normalized, their root fidelity is $\sqrt{F} = \sqrt{1 - r^2} = 1 - x/2$, and Eq. (25) gives $x_Q = x$. Thus the CHDF action coordinate has an exact finite A8b realization on $0 \leq x_A \leq 2$.

Existence is not physical selection. Composing the construction with a depolarizing channel replaces $r(x)$ by $\gamma r(x)$ for any $0 \leq \gamma \leq 1$. Every member is CPTP and preserves normalization, but

$$x_Q^{(\gamma)} = 2 \left(1 - \sqrt{1 - \gamma^2 r(x)^2} \right) \quad (27)$$

varies with γ . Positivity, normalization and CP descent therefore do not force the A8b member $\gamma = 1$. The complete-edge result of Paper III is a sufficient internal realization theorem once the corresponding edge is occupied; it does not prove that the physical base-seed parent selects that edge. The remaining source statement is precisely the occupation law $V_{\text{phys}} = V_{r(x)}$, or an independently derived bounded replacement for x_A . The range and counterfamily checks are reproduced by `audit_chdf_uhlmann_range_and_selection.py`.

Exact-recovery selector. The recovery branch removes the remaining freedom inside the depolarizing counterfamily if it is imposed on the full occupied qubit operator system. Let

$$\mathcal{C}_{\gamma}(\rho) = \gamma\rho + (1 - \gamma)I/2, \quad 0 \leq \gamma \leq 1. \quad (28)$$

For the two orthogonal eigenstates of σ_z , trace distance contracts from one to γ . If a CPTP map $\mathcal{R}$ satisfied $\mathcal{R}\mathcal{C}_{\gamma} = \text{id}$ on the full qubit code, trace-distance contractivity under $\mathcal{R}$ would imply

$$1 \leq D(\mathcal{C}_{\gamma}\rho_0, \mathcal{C}_{\gamma}\rho_1) = \gamma. \quad (29)$$

Hence $\gamma = 1$; conversely, $\gamma = 1$ has identity recovery. Exact full-code CPTP recoverability is therefore necessary and sufficient for the A8b member within this family. This is a genuine selector theorem, but not yet a Nature-selection theorem: the physical base-seed law has not been shown to require exact recovery on the full occupied operator system. Restricted state families and correctable subalgebras must not be silently promoted to that premise. The finite check is reproduced by `audit_depolarizing_recovery_selector.py`.

Protected logical recovery versus global loss. The phrase “full code” in the selector theorem refers to the full occupied logical ROOT operator system, not to every morphological-body realization degree of freedom. This distinction is required by the Tau architecture, whose observer descent is generically noninjective. Let

$$\mathcal{H}_B = \mathcal{H}_L \otimes \mathcal{H}_G, \quad \mathcal{E}_O = \text{Tr}_G, qquad \iota(\rho_L) = \rho_L \otimes \tau_G, \quad (30)$$

where L is the protected logical ROOT factor and G contains sourced realization multiplicity. Then

$$\mathcal{R}(\sigma_L) = \sigma_L \otimes \tau_G, qquad \mathcal{R}\mathcal{E}_O\iota = \iota, \quad (31)$$

although $\mathcal{E}_O$ is globally noninjective: states $\rho_L \otimes g_0$ and $\rho_L \otimes g_1$ with $g_0 \neq g_1$ have the same observer output. Thus exact recovery of the logical algebra is compatible with genuine observer-relative information loss in the body realization factor.

More generally, for an isometric body–observer–complement realization, exact recovery of the protected code is equivalent to the complementary output carrying no information about the logical state. This is the precise condition needed here: loss may retain realization or gauge information, but must be blind to the ROOT coordinate used by A8b. The construction proves compatibility and fixes the selector’s scope. It does not prove that the physical base–seed law places all ROOT information in the protected factor. That source-factorization statement is now the remaining physical premise. The finite logical/realization split is reproduced by `audit_protected_logical_recovery_split.py`.

ROOT–commutant factorization. The logical/realization split need not be introduced as an additional tensor postulate once one factor of the earlier Clifford packet has been physically designated as the logical ROOT factor on the occupied $M_8(\mathbb{C})$ block. Let

$$\mathcal{A}_L = M_2(\mathbb{C}) \otimes I_4. \quad (32)$$

Its commutant in $M_8(\mathbb{C})$ is

$$\mathcal{A}_G = \mathcal{A}'_L = I_2 \otimes M_4(\mathbb{C}). \quad (33)$$

The algebras commute, have complex dimensions four and sixteen, and their products span all 64 complex dimensions of M_8 . Hence a selected ROOT M_2 factor canonically determines the realization factor and the factorization $\mathcal{H}_B \simeq \mathcal{H}_L \otimes \mathcal{H}_G$ on this block. Equation (30) is therefore an internal consequence of the selected tensor packet rather than a further arbitrary enrichment.

This result does not remove the known tensor-selection boundary. Pointer records alone admit entangling conjugacies, and the current narrow parent does not select the required oriented Clifford flag. The remaining physical premise is now sharply the selection and occupation of the ROOT factor/flag; once supplied, its realization complement and protected-recovery split follow automatically. The finite commutant and spanning checks are reproduced by `audit_root_commutant_factorization.py`.

Factor-designation no-go. The rooted tensor packet does not itself perform that designation. Its root stabilizer S_3 permutes the three opposite-edge pairs, and therefore the three constructed M_2 factors, transitively. No individual factor projector is fixed by this action; the invariant subspace of the three-dimensional label representation is only the scalar line $(1, 1, 1)$. Thus OHC-P1 and the rooted TC-CVP1 germ conditionally select an oriented but unordered factor triple, not one distinguished logical ROOT factor.

Selecting one factor requires a typed S_3 -breaking datum. The weakest admissible form is a source-owned or irreducible observer–source access functional whose three factor weights are not all equal and whose unique extremum is frozen before terminal scoring. This datum may designate a logical access context without changing the body or adding a fourth factor. Its physical existence is not proved at this stage. Central chirality belongs to a separate odd extension and is later shown to be invisible to the current terminal. The permutation no-go is reproduced by `audit_root_factor_designation_no_go.py`.

Extremal observer-access selector. The preceding no-go can be sharpened without adding a terminal fit. After the three factor outputs have been source-typed into one common qubit codomain, consider the convex access family

$$\mathcal{C}_w = \sum_{k=1}^3 w_k \text{Tr}_{\hat{k}}, \quad w_k \geq 0, \quad \sum_{k=1}^3 w_k = 1, \quad (34)$$

where $\text{Tr}_{\hat{k}}$ retains factor k and traces out the other two. Encode a logical qubit in factor j while placing the other factors in the maximally mixed state,

$$\iota_j(\rho) = \rho_j \otimes I/2 \otimes I/2. \quad (35)$$

Direct substitution gives

$$\mathcal{C}_w \iota_j(\rho) = w_j \rho + (1 - w_j)I/2. \quad (36)$$

Thus the effective depolarizing parameter on the j th logical factor is $\gamma_j = w_j$. By the exact-recovery selector above, a CPTP left inverse on the full logical qubit algebra exists if and only if $w_j = 1$. Within Eq. (34), exact A8b recovery therefore selects an extremal simplex vertex $w = e_j$, not merely a unique largest weight.

This result supplies two useful controls. Symmetric access $w = (1/3, 1/3, 1/3)$ exactly designates no factor, while an interior vector with a unique maximum still cannot support exact full-algebra recovery. Under an S_3 relabeling, both e_j and the retained factor transform together, so different source-typed observer contexts may designate different factors without contradiction or alteration of the common body. The result is a conditional selector theorem, not by itself a derivation that Nature occupies a vertex. Qubit-terminal minimality and regular-event incidence below derive the corresponding on-code purity and contextual designation. The finite controls and permutation-covariance check are reproduced by `audit_extremal_observer_access_selector.py`.

Qubit-terminal minimality theorem. The vertex condition is not peculiar to the weighted family once the terminal type and recovery scope are fixed. Let the observer terminal be one full $M_2(\mathbb{C})$ algebra, and let the restriction of a CPTP access map $\mathcal{C}$ to one occupied logical qubit code admit a CPTP left inverse $\mathcal{R}$. Contractivity under both maps gives, for every pair of code states,

$$D(\rho, \sigma) = D(\mathcal{R}\mathcal{C}\rho, \mathcal{R}\mathcal{C}\sigma) \leq D(\mathcal{C}\rho, \mathcal{C}\sigma) \leq D(\rho, \sigma). \quad (37)$$

Hence the restriction is a trace-distance isometry. A reversible CPTP map between equal-dimensional full matrix algebras is unitary conjugation, so on the occupied code

$$\mathcal{C}(\rho) = U\rho U^\dagger. \quad (38)$$

It follows that exact full-code recovery through a minimal qubit terminal forces factor-pure access on that code; convex averaging or depolarization is excluded. Moreover, preserving two complete qubit factors jointly would require a four-dimensional code and cannot be reversibly represented in one two-dimensional terminal.

This removes factor purity as an independent premise after exact recovery and minimal terminal type have been adopted. It does not by itself select which member of the unordered factor triple is occupied, prove that the physical parent realizes exact recovery, or constrain off-code body information. Thus the remaining continuous datum at this stage is factor *designation*, not an additional gain or purity law. The dimension and scope audit is reproduced by `audit_qubit_terminal_minimality_selector.py`.

Observer–source commutant designation. The remaining factor label can be supplied relationally rather than by a preferred label in the common body. Let $\mathcal{N}_O$ be the algebra carried by the complementary observer output, equivalently the realization/noise algebra that is allowed to vary while the logical terminal is exactly recoverable. If this algebra is maximal on the realization complement of factor j , then

$$\mathcal{N}_O = I_2 \otimes M_4(\mathbb{C}), \quad \mathcal{A}_{\text{ROOT},O} = \mathcal{N}'_O = M_2(\mathbb{C}) \otimes I_4. \quad (39)$$

The double-commutant relation returns $\mathcal{N}_O$ and makes the designation unique for that typed context. Repeating the construction for each of the three factors gives three contexts in one S_3 orbit. Thus observers may designate different members of the same unordered body factorization while their descriptions remain related by the body’s permutation symmetry.

Equation (39) identifies what performs the designation: the observer–source complementary algebra, not a new body factor, fitted gain or globally preferred factor name. It is a conditional

source theorem. Exact recovery implies complementary blindness to logical information, but this commutant theorem alone does not force the *maximal* factor-complement algebra assumed here. A smaller noise algebra would have a larger commutant and would not uniquely select one qubit factor. The finite commutant dimensions and the three-context orbit are reproduced by `audit_observer_noise_commutant_selector.py`.

Canonical-expectation completion. Maximal complement occupation is not an additional free choice once a typed factor embedding, complete body occupation and the normalized trace are fixed. For

$$M_8(\mathbb{C}) = M_2(\mathbb{C}) \otimes M_4(\mathbb{C}), \quad (40)$$

the normalized-trace-preserving conditional expectation that is bimodular over the designated logical algebra is uniquely

$$\mathbb{E}_L = \text{id}_2 \otimes \tau_4, \quad (41)$$

where τ_4 is the normalized trace on the realization factor. Equivalently, the terminal channel is partial trace over that factor, up to the conventional normalization and identification of the logical output. Its minimal Kraus rank is four, so the complementary Hilbert space is four-dimensional and its full observable algebra is $M_4(\mathbb{C})$.

Thus the maximal complement in Eq. (39) follows from the already declared complete-body, trace-preserving and bimodular descent conditions after the logical embedding has been designated. The continuous source gap is reduced again: it is the typed observer–source incidence selecting the M_2 embedding itself. Non-bimodular, nonracial or incomplete-body maps lie outside this theorem and can have different complements. The dimension ledger is reproduced by `audit_canonical_expectation_complement.py`.

Regular-event incidence closes the factor label conditionally. The required typed incidence is not absent from the existing Tau completion. On the adopted E_1 /EREM observer branch, the earlier regular-event theorem supplies a nonzero compact event cochain j_{OS}^{evt} for every regular finite interaction. Resolve that cochain over the three factor sectors of the occupied tensor packet and define

$$w_k = \frac{\|P_k j_{OS}^{\text{evt}}\|^2}{\sum_{\ell=1}^3 \|P_\ell j_{OS}^{\text{evt}}\|^2}. \quad (42)$$

Regularity makes the denominator nonzero. The extremal-access and qubit-minimality theorems then imply that an exact full-qubit A8b terminal is possible only when $w = e_k$ for one unique k . That nonzero support identifies the typed M_2 embedding, after which the canonical expectation fixes the full complement.

No extra factor label or terminal fit has been introduced. A null event selects no factor. A regular event with support on several factor sectors is a valid observer interaction, but it cannot be promoted to an exact full-qubit A8b terminal; it belongs to a noisy, approximate or richer-terminal branch. Consequently the designation chain is closed conditionally on the adopted E_1 /EREM event completion and exact quantum recovery. The minimal parent still does not force either physical occupation premise. The odd chirality extension is outside this even-terminal selection. The null, mixed and three axis controls are reproduced by `audit_regular_event_factor_selection.py`.

Approximate-recovery stability. Exact recovery is the zero-error endpoint, not a requirement on every physical interaction. Adopt the explicit convention

$$\|\mathcal{R}\mathcal{C} - \text{id}\|_\diamond \leq \eta. \quad (43)$$

For two orthogonal logical states, each recovered state is then within trace distance $\eta/2$ of its target. The triangle inequality gives recovered-pair distance at least $1 - \eta$, while contractivity under $\mathcal{R}$

bounds it above by the pre-recovery distance. In the weighted access family that distance is w_j , hence

$$\boxed{w_j \geq 1 - \eta, \quad \sum_{k \neq j} w_k \leq \eta.} \quad (44)$$

Thus an experimentally bounded recovery error supplies a predeclared bound on off-factor incidence. At five-percent diamond error, for example, the selected factor carries at least 95% of the normalized access weight. Outside the weighted family the same argument bounds distinguishability contraction but does not by itself define unique factor weights. The finite error ledger is reproduced by `audit_approximate_factor_selection_bound.py`.

Central-chirality blindness of the present terminal. The central Cl_7 sign must be separated from the quantum terminal actually constructed here. Keep the six flag generators $\gamma_1, \dots, \gamma_6$ fixed and choose $\gamma_7 = \pm z_1 z_2 z_3$. The two extensions have opposite central Clifford volume, but the first six generators already generate all of $M_8(\mathbb{C})$. Therefore the three factor algebras, their states and effects, Born pairing, unitary transport, recovery maps and A8b geometry are identical as objects of the common M_8 terminal algebra.

Consequently central chirality is not a blocker for the even quantum-readout claim of Paper VI. It becomes physical only after a typed parity-odd, orientation-sensitive or Weyl terminal couples to the odd extension. Such a future terminal still requires a source-owned orientation lock, and the present blindness result neither selects its sign nor proves chirality to be gauge or absent. The two extensions and common rank-64 terminal algebra are reproduced by `audit_central_chirality_terminal_blindness.py`.

1.4 Predeclared readiness scoring

The alternative-source census can be ranked for protocol readiness without pretending to score a Tau signal. We froze a 13-point rubric before evaluating the candidates: same carrier (2), direct work/action-like response (2), endpoint geometry (2), loop holonomy (2), independent body-action bridge (3), reusable public numerical data (1), and errors/recovery (1). Partial capabilities receive one point only where the corresponding two-point item is physically present but incomplete. A hard gate overrides the total: eligibility for Δ_{AQ} scoring requires the same carrier, full endpoint and loop legs, and the independent three-point body-action bridge.

Table 2: Alternative-source readiness. Scores measure protocol completeness, not evidence for Tau Core.

Candidate	Score / 13	Hard Tau gate	Principal missing leg
Zhao Xmon geometric gate	8	fail	body action, reusable data
Cottet Maxwell demon	7	fail	holonomy, body action
Viyuela Uhlmann qubits	7	fail	direct work, body action
Batalhão NMR work	6	fail	holonomy, body action
Ghosh NMR mixed phase	6	fail	direct work, body action

The ranking selects the Xmon experiment as the closest integrated protocol architecture, while the Cottet and Viyuela experiments provide the strongest complementary work/tomography and holonomy realizations. The Tau-ready count is nevertheless zero. The score may guide source acquisition; it may not be converted into Δ_{AQ} , a significance, or a detection claim.

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